Lazy Prices in the S&P 100: A Replication with Dictionary and FinBERT Tone

Investment Research Portfolio · Project 15 · 10-K filings FY2012–2025, returns January 2014 – September 2026

Abstract

Cohen, Malloy and Nguyen (2020) show that US firms which barely change the wording of their annual and quarterly reports outperform firms that change them a lot. I repeat their core test on the current S&P 100 using 1,313 original 10-K filings for fiscal years 2012–2025. Each firm's Risk Factors (Item 1A) and MD&A (Item 7) are compared with the prior year's by cosine similarity. A monthly, equal-weighted portfolio that is long the most similar quintile and short the least similar earns 0.29% a month from January 2014 to September 2026 (t = 1.26; Sharpe 0.37). Its Fama-French five-factor plus momentum alpha is 0.20% a month, or 2.4% a year (t = 0.84, not significant). The value-weighted version earns an alpha of -0.33% (t = -1.23). In Fama-MacBeth regressions with size, momentum and short-term reversal controls, a one-standard-deviation increase in similarity predicts 0.21% higher next-month return (t = 2.89, significant at 5%). The equal-weighted direction matches the paper (the value-weighted one does not), but with about 78 large-cap firms a month the portfolio test has little power. Loughran-McDonald tone changes and FinBERT scores of 615,743 MD&A sentences add little.

1. Introduction

Cohen, Malloy and Nguyen (2020, *Journal of Finance*; NBER WP 25084) use every US 10-K and 10-Q filed from 1995 to 2014. They show that when a firm actively changes the language of its reports, the change predicts future earnings, profitability, news and even bankruptcies, yet there is no announcement effect: returns accrue only when the information is later revealed, suggesting investors are inattentive. Their main measure is the similarity of a filing's text to the same filing a year earlier. From 1995 to 2014 the most similar quintile beat the least similar by 0.31% a month (t = 3.13) equal-weighted, with a five-factor alpha of 0.32% (t = 4.21).

This replication asks a narrower question that matters to a practitioner: does the effect show up after the paper's sample ends, in the largest and most closely followed US companies? Large firms' filings are read by more analysts, which should make 'lazy' pricing less likely. The test also adds two tone measures, the Loughran-McDonald (2011) word lists and FinBERT (Araci 2019; ProsusAI/finbert), to see whether what a filing says adds to how much it changed.

2. Data

Universe. The 101 current S&P 100 constituents (Wikipedia list, 7 October 2026; Alphabet's two share classes are one firm, so 100 firms). **Filings.** All 10-Ks for fiscal years 2011–2025 were downloaded from SEC EDGAR with the *edgartools* package. For each firm and fiscal year only the original filing is used (amendments are dropped), and Item 1A and Item 7 are extracted. When edgartools' parser returns a short or empty item (under 5,000 characters for Item 1A or 8,000 for Item 7, typically a cross-reference), the section is taken from the full 10-K text by regular expression or, where the 10-K incorporates it by reference, from the Exhibit 13 annual report. **Returns.** Monthly total returns are from yfinance adjusted closes. Market capitalisation = shares outstanding from the XBRL cover page (summed over share classes) × month-end close. Fama-French five factors and momentum come from Kenneth French's data library. **Dictionary.** Loughran-McDonald Master Dictionary 1993–2025 (Notre Dame SRAF).

Section taken from	Item 1A	Item 7
10-K item	1252	1113
10-K regex	2	9
exhibit 13	42	78
exhibit 13 (capped)	0	3
missing	17	110

Table 1. Where each section's text comes from, all original 10-Ks FY2012–2025.

Usable pairs. A similarity score is used only if both years' sections come from the 10-K itself, by the same extraction route, for consecutive fiscal years. Comparing an Exhibit 13 extract with a 10-K item would measure formatting rather than disclosure. This leaves **1,067 firm-year pairs from 94 firms**. The largest exclusions are banks that publish MD&A in Exhibit 13, and firms that were not yet public or filed 20-Fs early in the sample.

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
10-Ks (original filings)	88	88	88	90	91	92	94	96	96	97	97	97	99	100
Usable similarity pairs	55	64	62	67	72	70	72	78	82	86	88	89	90	92

Table 2. Coverage by fiscal year.

3. Method

- **Cleaning.** Lines with more digits than letters (table rows) are dropped, text is lowercased, only alphabetic tokens are kept (removing numbers), and English stop words (scikit-learn list) and one-letter tokens are removed.
- **Similarity.** Cosine similarity of word-count vectors between each 10-K and the prior year's, separately for Item 1A, Item 7 and the two combined (the main signal). A higher value means fewer changes.
- **Dictionary tone.** Share of all words (before stop-word removal) in the Loughran-McDonald negative, uncertainty and litigious lists, and the year-over-year change in each.
- **FinBERT tone.** Item 7 is split into sentences of at least five words (615,743 in total). Each is scored by ProsusAI/finbert (maximum 256 tokens), and the firm-year score is the mean of P(positive) − P(negative).
- **Portfolios.** At each month-end from December 2013 to August 2026, every firm carries its most recent usable signal filed on or before that date and at most 15 months old. Firms are ranked into similarity quintiles (about 16 stocks each). The strategy is long Q5 (most similar) and short Q1 (least similar), equal-weighted, rebalanced monthly. Returns run from January 2014 to September 2026.
- **Statistics.** Mean, Sharpe ratio and t-statistic (Newey-West, 6 lags), and alpha from a regression on the Fama-French five factors plus momentum.
- **Fama-MacBeth.** Each month, next-month return (%) is regressed cross-sectionally on similarity, the change in the negative-word share and the FinBERT score, controlling for log market cap, 12-1 month momentum and the prior-month return. Coefficients are averaged with Newey-West t-statistics. Specification (a) uses raw variables, comparable with the paper's Table IV. Specification (b) standardises each regressor every month, so each coefficient is the return per one standard deviation.

4. Results

4.1 How much do S&P 100 10-Ks change?

	count	mean	std	1%	50%	99%
Similarity, Item 1A	1,067	0.9782	0.0445	0.7935	0.9885	0.9993
Similarity, Item 7	1,067	0.9700	0.0223	0.8893	0.9767	0.9952
Similarity, combined	1,067	0.9789	0.0193	0.9106	0.9845	0.9972
LM negative share	1,067	0.0259	0.0056	0.0130	0.0255	0.0386
Δ LM negative share	1,067	0.0006	0.0016	-0.0034	0.0005	0.0050
FinBERT MD&A tone	1,067	0.0470	0.0705	-0.1271	0.0454	0.2098

Table 3. Descriptive statistics of the usable pairs. The paper's full-sample cosine similarity has a median of 0.947 and an SD of 0.191 (Table I, Panel B).

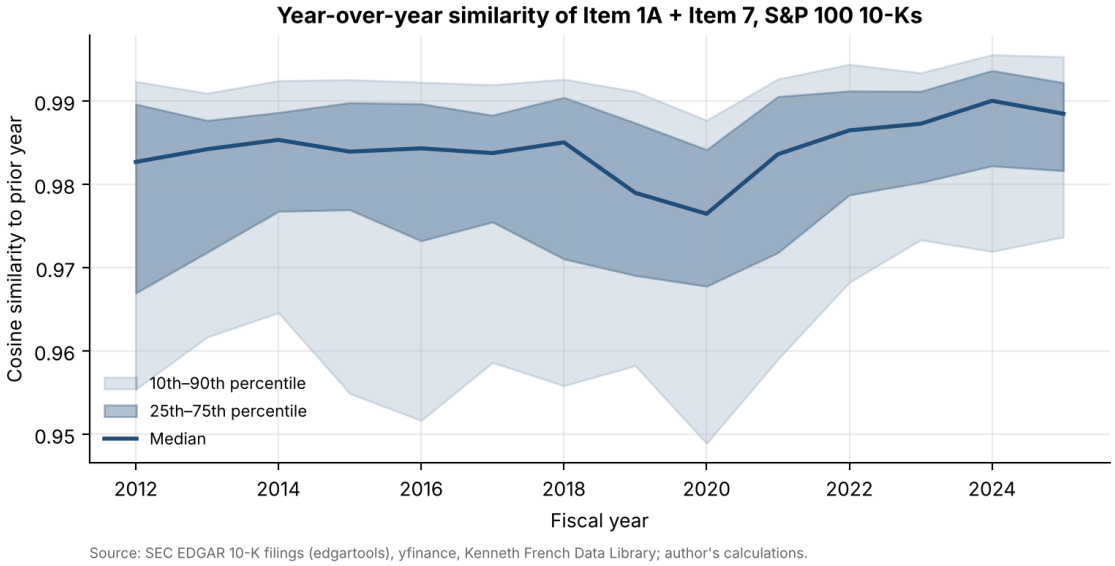


Figure 1. Distribution of year-over-year similarity (Items 1A + 7) by fiscal year.

S&P 100 filings are even more repetitive than the paper's full sample: the median combined similarity is 0.985, and the 1st percentile is 0.911. Big changes are rare. The five lowest scores in the sample are AMAT FY2023 (0.79), TMUS FY2013 (0.83), SCHW FY2017 (0.83), DELL FY2021 (0.84), GE FY2016 (0.86) (the ten lowest are in data/largest_changers.csv). Similarity is positively but not strongly related to tone: the Spearman correlation between combined similarity and the change in the negative-word share is -0.15, and with FinBERT tone 0.01.

Ticker	Fiscal year	Filed	Similarity 1A	Similarity 7	Similarity combined
AMAT	2023	2023-12-15	0.784	0.843	0.792
TMUS	2013	2014-02-25	0.867	0.805	0.826
SCHW	2017	2018-02-22	0.660	0.886	0.828
DELL	2021	2021-03-26	0.505	0.965	0.844
GE	2016	2017-02-24	0.504	0.901	0.864
GE	2021	2022-02-11	0.959	0.861	0.882
NVDA	2012	2012-03-13	0.643	0.976	0.892
MO	2022	2023-02-27	0.845	0.897	0.893
MU	2014	2014-10-27	0.920	0.881	0.899
PM	2018	2019-02-07	0.981	0.889	0.902

Table 4. The ten largest year-over-year changes among the usable pairs.

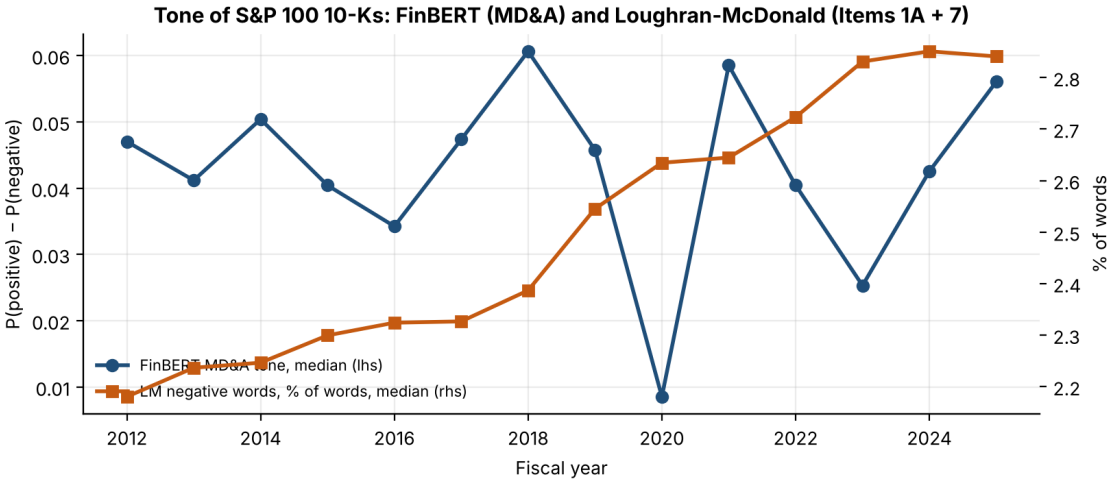


Figure 2. Median tone by fiscal year.

4.2 Portfolio test

	Excess return / month	t	Sharpe (ann.)	Alpha FF5+Mom / month	t(alpha)
Q1 (changers)	1.30%	4.85	1.00	0.34%	2.18
Q2	1.42%	4.54	1.03	0.39%	2.20
Q3	1.61%	6.35	1.27	0.63%	5.60
Q4	1.52%	5.66	1.20	0.61%	4.72
Q5 (non-changers)	1.58%	4.59	1.10	0.53%	4.04
Q5 – Q1	0.29%	1.26	0.37	0.20%	0.84

Table 5. Similarity quintile portfolios, equal-weighted, January 2014 – September 2026 (153 months). Q1–Q5 returns are in excess of the risk-free rate; Q5 – Q1 is a zero-cost spread. Newey-West t-statistics.

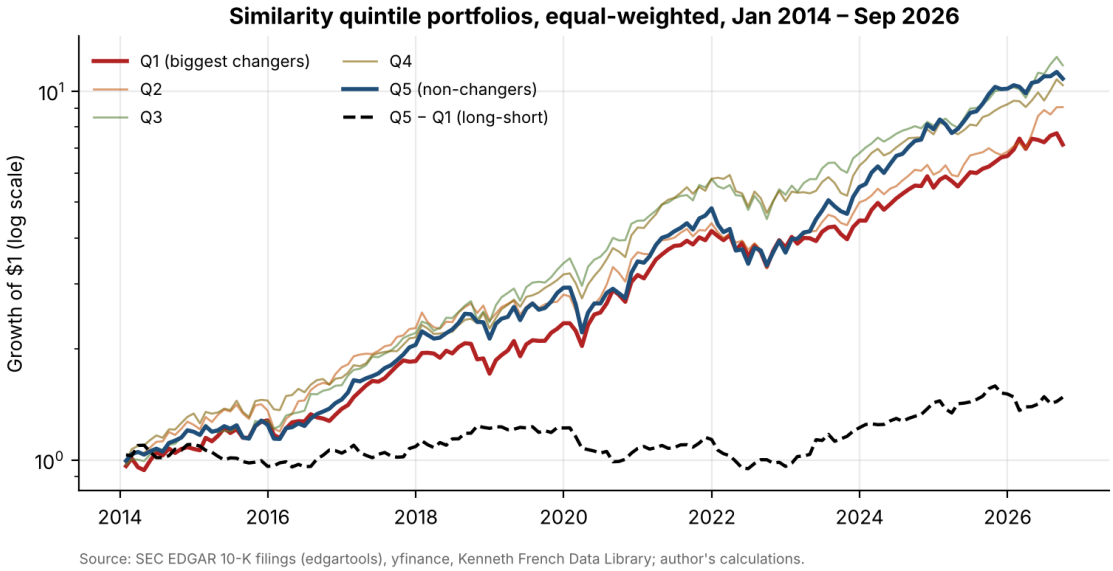


Figure 3. Cumulative returns of the quintile portfolios and the long-short spread.

The spread is positive, as in the paper: non-changers beat changers by 0.29% a month, or 3.1% a year compounded, with an annualised volatility of 9.5%. But it is not significant (t = 1.26). After the six factors, the alpha is 0.20% a month (t = 0.84). The factor loadings show what the spread is exposed to: market 0.10, size -0.09, value 0.06, profitability -0.24, investment -0.28, momentum -0.07 (R^2 0.14). Figure 4 shows the alpha of each quintile.

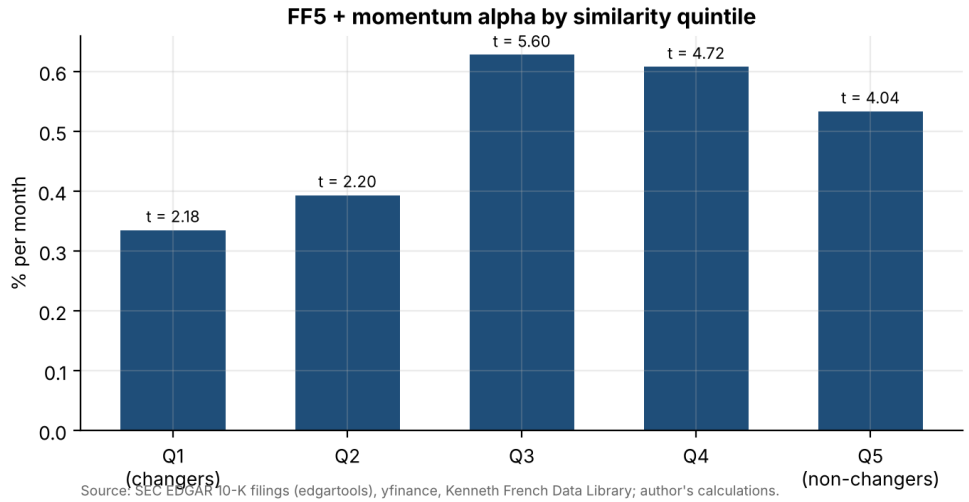


Figure 4. Six-factor alpha by similarity quintile.

	Q5 − Q1 / month	t	Sharpe	Alpha / month	t(alpha)	Months
Items 1A + 7 · equal-weight	0.29%	1.26	0.37	0.20%	0.84	153
Item 1A only · equal-weight	0.29%	1.47	0.37	0.17%	0.88	153
Item 7 only · equal-weight	-0.14%	-0.44	-0.15	0.05%	0.17	153
Items 1A + 7 · value-weight	-0.34%	-1.14	-0.34	-0.33%	-1.23	153
Items 1A + 7 · 2014–2019	0.31%	1.21	0.44	0.29%	1.12	72
Items 1A + 7 · 2020–2026	0.28%	0.74	0.32	0.17%	0.45	81
Items 1A + 7 · terciles	0.07%	0.38	0.10	0.08%	0.45	153
Items 1A + 7 · sector-adjusted	0.29%	1.18	0.35	0.26%	1.21	153

Table 6. Robustness of the long-short spread: section, weighting, sub-period and terciles (about 30 stocks a side) and a sector-adjusted signal (similarity minus its GICS sector's median each month).

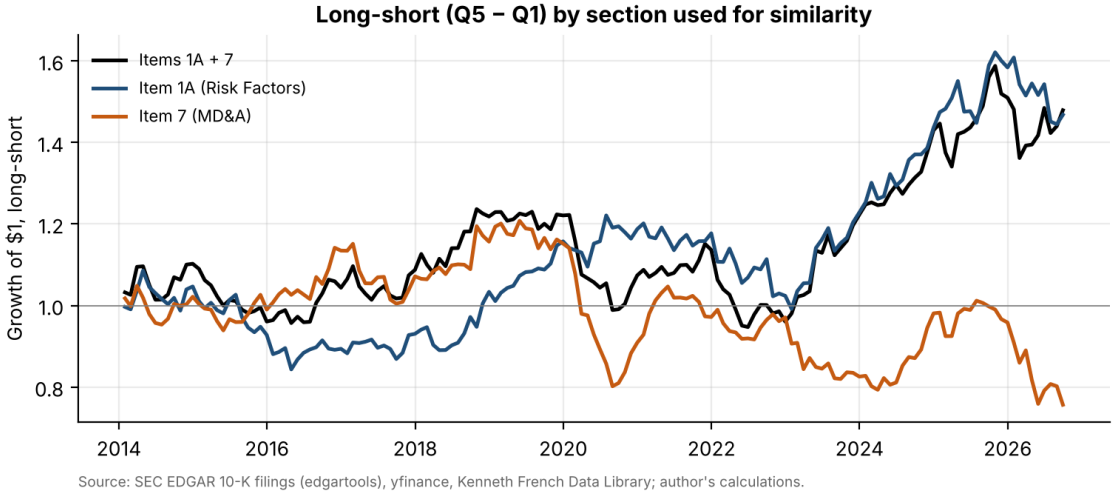


Figure 5. Long-short growth of $1 using each section's similarity.

Every quintile has a positive six-factor alpha, from 0.34% (Q1) to 0.63% a month. That is the footprint of survivorship: today's S&P 100 is, by construction, a list of firms that did well. The long-short spread nets this out, which is why it is the right test statistic. The spread comes from Item 1A (Risk Factors), 0.29% a month, while Item 7 alone gives -0.14%. This matches the paper's finding that changes to the risk-factor section are especially informative. The result is similar in 2014–2019 and 2020–2026, and the sector-adjusted signal earns 0.29% (t = 1.18). None of the variants is significant at 5%.

	Q1	Q2	Q3	Q4	Q5
Firms per month	15.9	15.2	15.4	15.2	15.8
Market cap ($bn), median	108	120	134	133	153
Similarity (Items 1A + 7)	0.951	0.977	0.984	0.988	0.993
Δ LM negative share (pp)	0.096	0.064	0.053	0.045	0.032
FinBERT MD&A tone	0.049	0.038	0.056	0.050	0.043
Financials, share of firms	7%	6%	10%	19%	25%
Health Care, share of firms	19%	16%	20%	20%	9%
Information Technology, share of firms	19%	25%	23%	26%	31%

Table 7. Characteristics of the similarity quintiles, averaged over the 153 formation months (compare the paper's Table III). Sectors are today's GICS sectors (Wikipedia list).

As in the paper's Table III, where the tone of the changed text is four times as negative in Q1 as in Q5, changers' language turns more negative: the negative-word share rises by 0.096pp a year in Q1 against 0.032pp in Q5. Unlike the paper, where changers are the larger firms, changers here are smaller (median $108bn vs $153bn). The sector mix differs sharply: 25% of non-changers are financials against 7% of changers, while health care is 19% of changers and 9% of non-changers. That is why the sector-adjusted variant matters: it gives 0.29% a month against 0.29% for the raw signal.

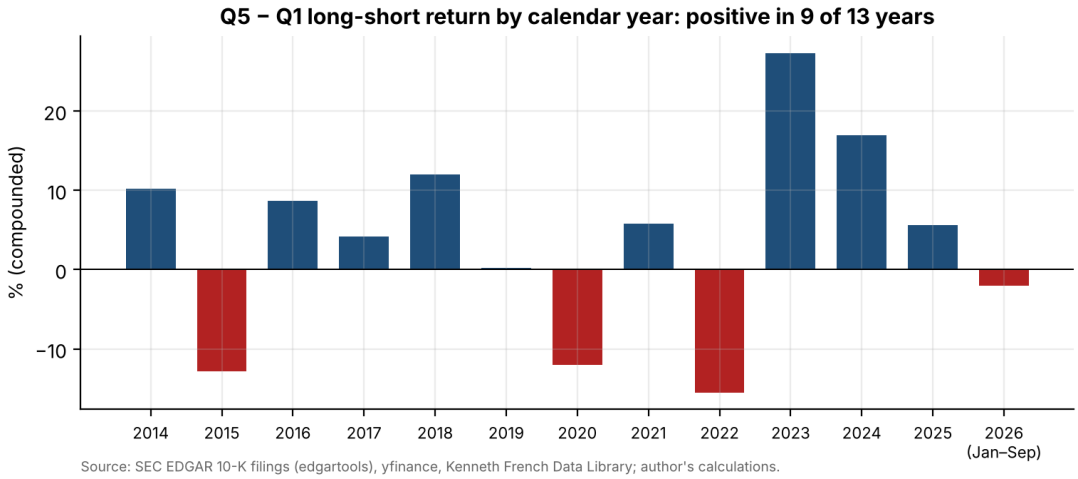


Figure 6. Long-short return by calendar year; positive in 9 of 13 years.

4.3 Fama-MacBeth regressions

	(1) Similarity	(2) + controls	(3) + tone	(4) Tone only
Intercept	1.600 (6.22)	1.605 (6.19)	1.605 (6.19)	1.536 (5.99)
Similarity (Items 1A + 7)	0.156 (2.17)	0.207 (2.89)	0.230 (3.17)	
Δ LM negative share			0.087 (0.89)	-0.013 (-0.14)
FinBERT MD&A tone			0.033 (0.32)	0.011 (0.12)
log market cap		-0.369 (-3.95)	-0.402 (-4.42)	-0.363 (-4.07)
Momentum (12-1)		0.542 (3.21)	0.512 (3.10)	0.523 (3.21)
Prior-month return		0.035 (0.28)	0.015 (0.12)	0.003 (0.02)
Months	153	153	153	153
Avg firms	78	75	75	83
Avg adj. R^2	0.002	0.100	0.107	0.101

Table 8. Fama-MacBeth regressions of next-month return (%), standardised regressors: coefficient = % return per one cross-sectional SD. Newey-West t-statistics in parentheses.

	(1) Similarity	(2) + controls	(3) + tone	(4) Tone only
Intercept	-5.637 (-1.39)	0.530 (0.12)	0.469 (0.10)	11.047 (4.14)
Similarity (Items 1A + 7)	7.396 (1.77)	10.852 (2.64)	11.814 (2.85)	
Δ LM negative share			66.408 (1.27)	2.353 (0.08)
FinBERT MD&A tone			0.326 (0.22)	0.196 (0.16)
log market cap		-0.395 (-3.82)	-0.431 (-4.36)	-0.391 (-3.94)
Momentum (12-1)		1.462 (3.47)	1.360 (3.21)	1.449 (3.44)
Prior-month return		-0.036 (-0.02)	-0.293 (-0.16)	-0.534 (-0.30)
Months	153	153	153	153
Avg firms	78	75	75	83
Avg adj. R^2	0.002	0.100	0.107	0.101

Table 9. Fama-MacBeth regressions of next-month return (%), raw regressors, comparable with the paper's Table IV (Sim_Cosine 0.31, t = 2.51). Newey-West t-statistics in parentheses.

With controls, a one-SD increase in similarity predicts a 0.21% higher next-month return (t = 2.89; significant at 5%). In raw units the coefficient is 10.85 (t = 2.64), against 0.31 in the paper. Because S&P 100 similarity varies much less than in the paper (SD 0.019 vs 0.191), the raw coefficients are not directly comparable, and the standardised effect is the better comparison: 0.31 × 0.191 ≈ 0.06% a month per SD in the paper. Adding tone (specification 3): Δ negative share 0.09 (t = 0.89), FinBERT tone 0.03 (t = 0.32), and similarity stays at 0.23 (t = 3.17).

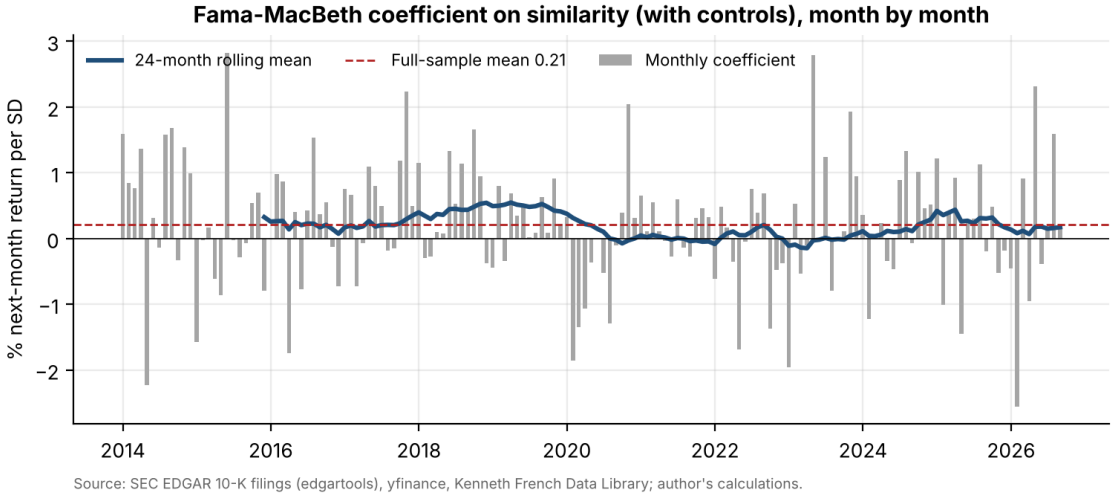


Figure 7. Monthly Fama-MacBeth coefficient on standardised similarity (specification 2).

The monthly coefficients are noisy, which is typical of Fama-MacBeth estimates with about 75 firms. The average is positive, and the rolling mean shows how much of it depends on a few periods.

4.4 Do the tone measures sort returns on their own?

	Q5 − Q1 / month	t	Alpha / month	t(alpha)
Δ LM negative share (Q5 = biggest increase)	0.17%	0.62	0.23%	1.13
FinBERT MD&A tone (Q5 = most positive)	0.09%	0.29	0.03%	0.11
Δ FinBERT tone (Q5 = biggest improvement)	0.06%	0.22	-0.05%	-0.24

Table 10. Long-short quintile spreads sorted on tone, equal-weighted, January 2014 – September 2026.

Sorted on its own, neither tone measure produces a reliable spread: the largest t-statistic in Table 10 is 0.62. In the Fama-MacBeth regressions FinBERT tone has a positive but insignificant coefficient, and adding both tone measures does not weaken the similarity coefficient. Similarity therefore carries information that the level and change of tone do not, as the paper argues.

4.5 Is there an announcement effect?

	Q5 (non-changers)	Q1 (changers)	Q5 – Q1	t	Filings per quintile
Month 0 (filing month)	0.26%	0.95%	-0.68%	-0.88	215
Months 1–3	-0.88%	0.75%	-1.63%	-1.24	215
Months 4–6	1.12%	-0.44%	1.57%	1.51	215
Months 7–12	1.83%	-1.01%	2.84%	1.66	215

Table 11. Cumulative abnormal return (stock minus the equal-weighted sample average) in event-time windows after the 10-K filing, for filings in the most and least similar quintile of their fiscal-year cohort (filings from 2013). t-statistics treat filings as independent.

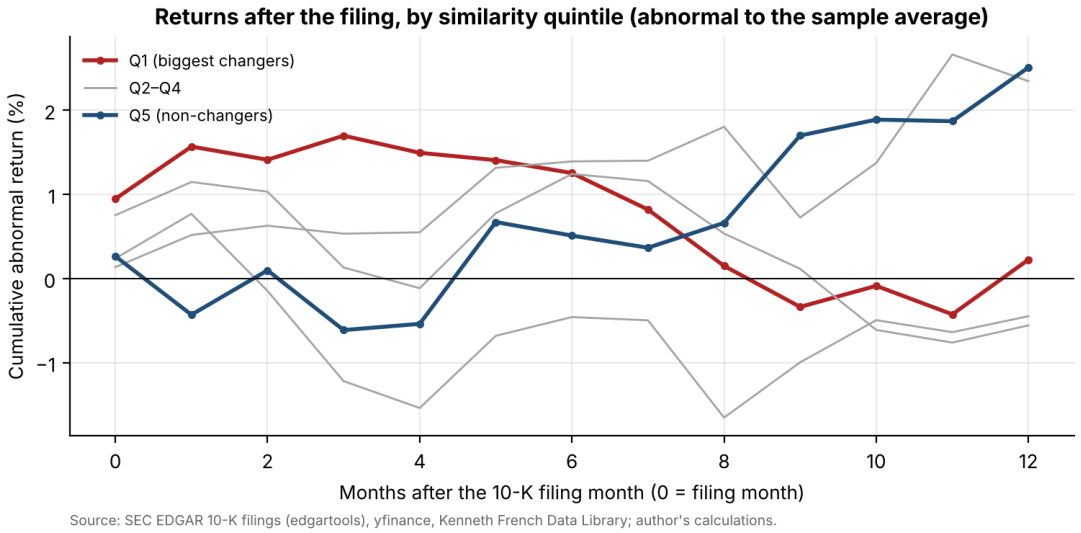


Figure 8. Cumulative abnormal return after the filing month, by similarity quintile.

The paper's most striking claim is that the market does not react when a changed 10-K is filed, and the return gap opens later as the news comes out. The S&P 100 data fit that pattern: in the filing month and the next three months changers do no worse than non-changers (Q5 – Q1 = -0.68% and -1.63%, t = -0.88 and -1.24). The gap opens in months 4–6 (1.57%, t = 1.51) and 7–12 (2.84%, t = 1.66). With about 215 filings per quintile and overlapping windows, this is suggestive rather than conclusive.

5. Comparison with Cohen, Malloy and Nguyen (2020)

	Cohen, Malloy & Nguyen (2020)	This replication
Sample	All US firms with 10-K/10-Q text, 1995–2014	Current S&P 100 (94 firms with usable pairs), returns Jan 2014 – Sep 2026
Documents	86,965 10-Ks + 258,271 10-Qs	1313 original 10-Ks (FY2012–2025)
Text	Full 10-K and 10-Q	Item 1A + Item 7 of the 10-K
Holding	Enter the month after filing, hold 3 months	Most recent 10-K signal, re-ranked monthly
Factor model for alpha	Market, size, value, momentum, liquidity	Fama-French 5 + momentum
Median cosine similarity	0.947	0.985
EW Q5–Q1 excess return (monthly)	0.31% (t = 3.13)	0.29% (t = 1.26)
EW Q5–Q1 alpha (monthly)	0.32% (t = 4.21)	0.20% (t = 0.84)
VW Q5–Q1 alpha (monthly)	0.34% (t = 2.53)	-0.33% (t = -1.23)
Fama-MacBeth coefficient on similarity (with controls)	0.31 (t = 2.51)	10.85 (t = 2.64)
Same, per one standard deviation of similarity	≈0.06% a month (0.31 × SD 0.191)	0.21% a month (t = 2.89)

Table 12. The original paper (NBER WP 25084, March 2019 revision: Table I Panel A/B, Table II Sim_Cosine, Table IV column 2) and this replication.

The direction is the same as in the paper: non-changers outperform changers. The size of the effect is in the same range (0.29% vs 0.31% a month), but the statistical strength is much weaker. That is what one would

expect with about 78 firms instead of several thousand, 153 months instead of the paper's 20 years, and a universe of mega-caps. The paper's value-weighted result (0.34%, t = 2.53) shows the effect is not confined to small stocks; here the value-weighted spread is -0.34% (t = -1.14), the opposite sign, in a universe where a few very large firms carry most of the weight.

6. Limitations

- **Sample size.** About 78 firms with a usable signal each month means about 16 stocks per quintile, so each stock is about 6% of its leg, so the test cannot detect an effect of the paper's size (0.3% a month) with confidence. The original uses all US firms with EDGAR filings (86,965 10-Ks and 258,271 10-Qs).
- **Survivorship and look-ahead bias.** The universe is today's S&P 100, so firms are included because they became large and survived to 2026. Earlier members that shrank, were acquired or failed are missing, and some current members were small in 2014. This biases average returns up and may weaken the changer effect, because the failures that the paper links to big changes are excluded.
- **10-K only, once a year.** The paper also uses 10-Qs and holds each signal for three months after filing. With 10-Ks only, the signal is up to a year old, and the portfolio changes little from month to month.
- **Section extraction.** Items are parsed by edgartools, with regex and Exhibit 13 fallbacks. Pairs that mix extraction routes are excluded rather than corrected, which removes most banks' MD&A. Parsing errors would show up as false 'changers'.
- **Factor model.** Fama-French five factors plus momentum, not the paper's market, size, value, momentum and liquidity model. The two are similar but not identical.
- **Tone models.** FinBERT was trained on financial news and analyst reports, not 10-Ks, and truncates sentences at 256 tokens. The Loughran-McDonald lists are counted without the paper's 'sentiment of change' (tone of only the changed text).
- **Costs.** Returns are before trading costs. The long-short portfolio is cheap to trade because signals change once a year.

References

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